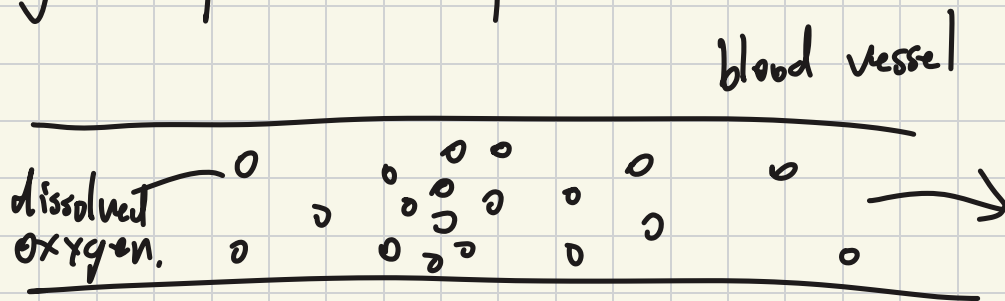


Lecture 4

Review + finish Picard's iteration, see lecture 3.

Method of characteristics

Motivating example + warmup



$u(t, x) = (\text{mass of dissolved oxygen}) / (\text{unit length})$ at (t, x)

total flux interval $[a, b]$

$$m(t) := \int_a^b u(t, x) dx = \text{mass of DO on } [a, b]$$

$v(t, x) = \text{velocity}$, flux $q(t, x) = v(t, x)u(t, x)$.

$$\leadsto \frac{dm}{dt}(t) = q(t, a) - q(t, b) = - \int_a^b \frac{\partial q}{\partial x} dx$$

$$\leadsto \int_a^b \frac{du}{dt} dx = - \int_a^b \frac{\partial q}{\partial x} dx \quad \forall a, b$$

$$\Rightarrow \frac{du}{dt} + v \frac{\partial u}{\partial x} + u \frac{\partial v}{\partial x} = 0$$

Consider equation of the form

$$\frac{\partial u}{\partial t} + v \frac{\partial u}{\partial x} + w = 0$$

(*)

$v = v(t, x)$ independent of u

$w = w(t, x, u)$

Def. $t \mapsto x(t)$ is called a characteristic if

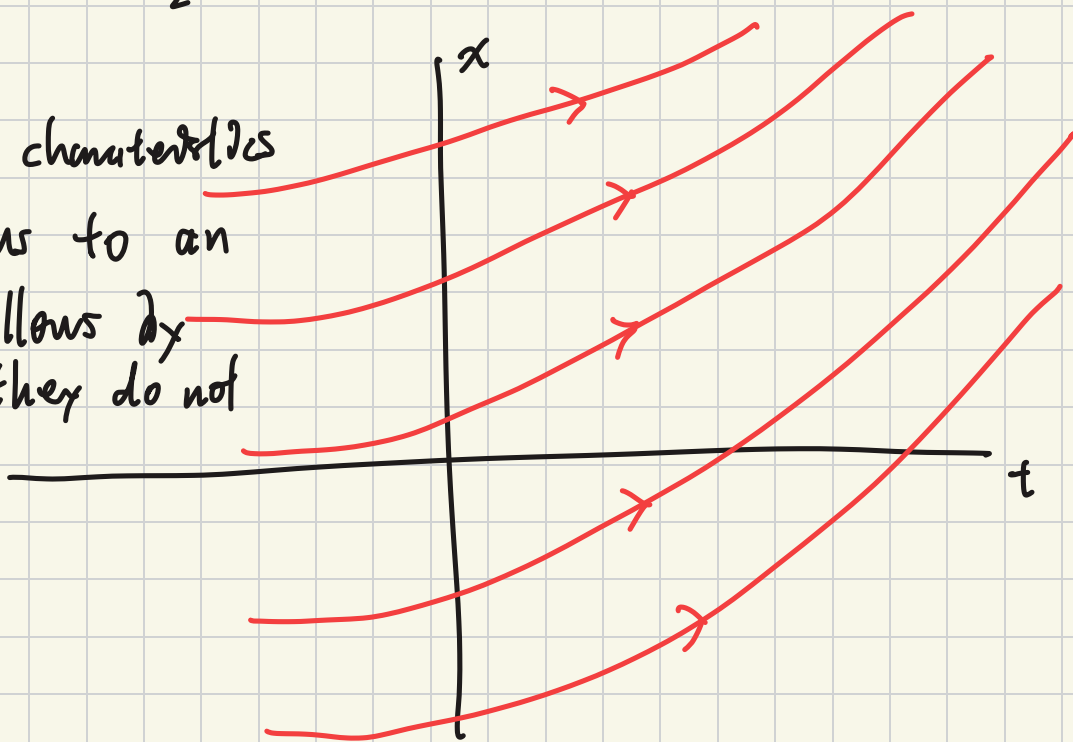
$$\frac{dx}{dt}(t) = v(t, x(t))$$

Recall if v is sufficiently nice, say C^1 , then $\exists$ unique solution in a nbhd of any starting point (t_0, x_0) .

Ex. If $v(t, x) = at + b$, the characteristics are

$$x(t) = \frac{a}{2}t^2 + bt + x_0$$

Rmk. Since characteristics are solutions to an ODE, it follows by E+U that they do not intersect.



Thm. Assume u solves $\textcircled{*}$ and $x(t)$ is a characteristic, then $u(t, x(t))$ satisfies the ODE

$$\frac{d}{dt} u(t, x(t)) = -w(t, x(t), u(t, x(t))).$$

Pf. Apply chain rule:

$$\begin{aligned} \frac{d}{dt} u(t, x(t)) &= \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} \frac{\partial x}{\partial t} \\ &= -v \frac{\partial u}{\partial x} - w + \frac{\partial u}{\partial x} \cdot v \\ &= -w \end{aligned} \quad \square$$

Interpretation: Along trajectories given by the velocity, the oxygen concentration satisfies an ODE.

Idea: solve $\textcircled{*}$ by solving ODEs along characteristics.

Ex 0.
$$\begin{cases} \partial_t u + a \partial_x u = 0 \\ u(0, x) = g(x) \end{cases}$$

Characteristics: $\frac{dx}{dt} = a \Rightarrow x(t, s) = at + s \quad s \in \mathbb{R}$

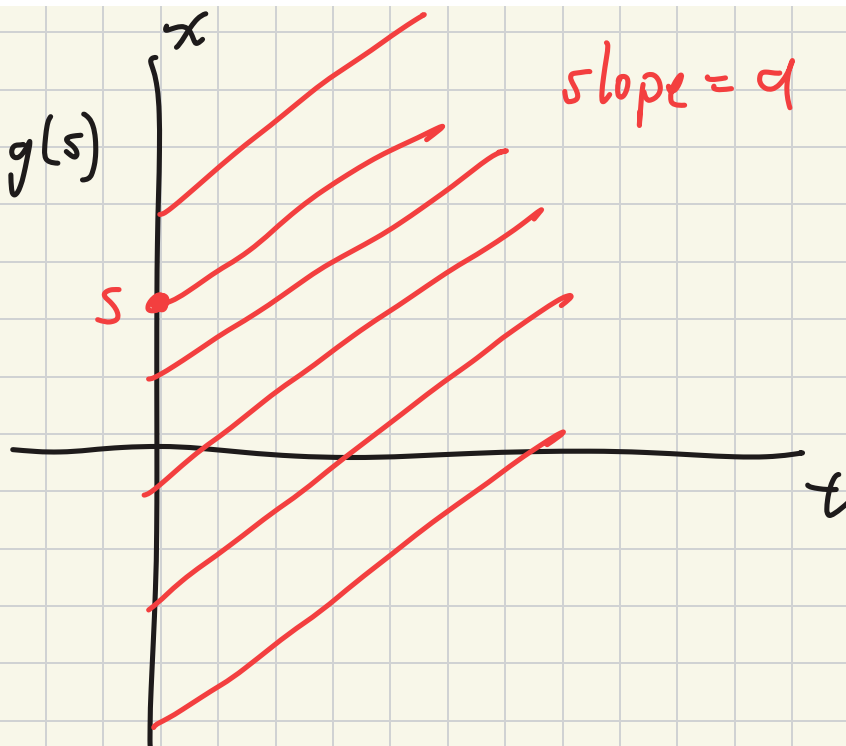
by previous Thm, u is constant along characteristics

$$u(t, at+s) = u(0, s) = g(s)$$

$$x = at + s$$

$$s = x - at$$

$$u(t, x) = g(x - at)$$



Ex 1. Let $a, b \in \mathbb{R}$, $u(t, x)$ satisfies

$$\begin{cases} \frac{\partial u}{\partial t} + (at+b) \frac{\partial u}{\partial x} = 0 \\ u(0, x) = g(x) \end{cases}$$

Characteristics $x(t, s) = \frac{a}{2} t^2 + b t + s$

Note that $x(0, s) = s$

$$\frac{d}{dt} u(t, \frac{a}{2} t^2 + b t + x_0) = 0$$

$$\begin{cases} \frac{d}{dt} u(t, x(t, s)) = 0 \\ u(0, x(0, s)) = g(s) \end{cases}$$

$$\Rightarrow u(t, \frac{a}{2}t^2 + bt + s) = u(0, s) = g(s)$$

Want formula for $u(t, x)$, so identify

$$x = \frac{a}{2}t^2 + bt + x_0$$

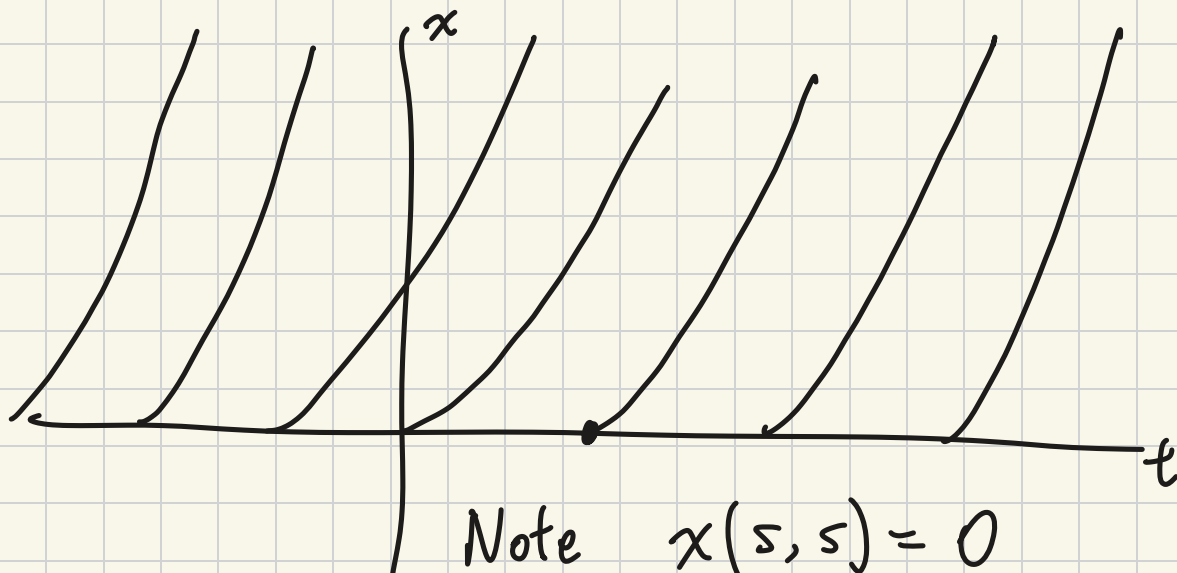
$$\leadsto u(t, x) = g(x - \frac{a}{2}t^2 - bt).$$

Ex 2 velocity depending on position instead of time.

$$\begin{cases} \frac{\partial u}{\partial t} + (a + bx) \frac{\partial u}{\partial x} + bu = 0 \\ u(t, 0) = f(t) \end{cases}$$

Characteristic eq. $\frac{dx}{dt} = a + bx$, solutions

$$x(t, s) = \frac{a}{b} [e^{b(t-s)} - 1] \quad \text{indexed by } s$$



$$\begin{cases} \frac{d}{dt} u(t, x(t, s)) + b u(t, x(t, s)) = 0 \\ u(t, x(t, s)) \Big|_{t=s} = f(s) \end{cases}$$

general sol

$$u(t, x(t, s)) = C_s e^{-bt}$$

with initial condition

$$u(t, x(t, s)) = f(s) e^{-b(t-s)}$$

$$u\left(t, \frac{a}{b} [e^{b(t-s)} - 1]\right) = f(s) e^{-b(t-s)}$$

$$\text{put } x = \frac{a}{b} (e^{b(t-s)} - 1)$$

$$\leadsto s = t + \frac{1}{b} \log \frac{a}{a+bx}$$

$$\Rightarrow u(t, x) = \frac{a}{a+bx} f\left(t + \frac{1}{b} \log \frac{a}{a+bx}\right)$$

Concluding remarks. Characteristics gives rise to a change of variables. A family of characteristics $x(t, s)$ can be viewed as a map

$$(t, s) \mapsto (t, x(t, s))$$

In (t, s) coordinates, PDE is easy to solve, then need to change variables back to (t, x) .